

FROM 0% TO 90%

Systematic RAG Retrieval Optimization

A case study in data-driven engineering: How 8 focused sessions and 7.5 hours of systematic work transformed a broken search system into elite-tier accuracy.

90%

ACCURACY

8

SESSIONS

100%

ARCHITECTURAL

100%

COMPARATIVE

ROOT CAUSE ANALYSIS

THE FUNDAMENTAL DIAGNOSIS

Analysis identified three distinct technical failures preventing accurate retrieval. Vector search alone finds "related" noise instead of "authoritative" signals.



KEYWORDS

Vector-only search misses exact term matches. Query: "What is Nomad?" result in semantic drift. Semantic $\neq$ Exact.



AUTHORITY

Session notes rank equally with authoritative ADRs. Frequency of mentions outranked structural definitions.



TOPIC MATCH

Docs that mention a topic outrank docs primary ABOUT a topic. The "About" signal was buried in chunk frequency.

THE THREE PILLARS STRATEGY

Systematic optimization through three complementary techniques, validated with a 20-query test suite. Every pillar addresses a root cause.



PILLAR 1: HYBRID

BM25 (Keywords) + Vector (Semantics). Fixed the exact-match failure.

Result: 0% → 20%



PILLAR 2: AUTHORITY

ADR boosting (1.5×) and noise de-boosting (0.8×). Rank definitions first.

Result: 20% → 75%



PILLAR 3: TOPIC

Title and filename matching boost (1.5×). Ensure "About X" outranks "Mentions X".

Result: 75% → 90%

BREAKTHROUGH #1: HYBRID SEARCH

THE TECHNICAL FORMULA

$$\text{Score} = 0.3 \times \text{BM25} + 0.7 \times \text{Vector}$$

Custom tokenizer preserves technical identifiers like llama3.1:8b and ADR-001 as single atoms rather than split fragments.

THE OUTCOME

- Vector Only: "What is Nomad?" finds casual session mentions of the cluster.
- Hybrid: Exact "Nomad" keyword forces the definition ADR to the top rank.

Foundational Validation Complete.

BREAKTHROUGH #2: DOCUMENT AUTHORITY

Insight: Not all documents are equal. ADRs are authoritative definitions; Session notes are casual noise. We measured the gap and implemented data-driven boosting.

BOOST MULTIPLIERS

ADR: 1.5× (Definition)

Reports: 1.3× (Formal)

Sessions: 0.8× (De-boost)

THE 55% JUMP

Gap analysis showed ADR-001 needed exactly 1.5× to clear the noise. Calculated, not guessed.

Result: 20% → 75%

BREAKTHROUGH #3: TOPIC AUTHORITY

SOLUTION: TITLE MATCHING

Resolved "Semantic Competition" where multiple docs mention a query term. Boost documents where terms appear in the TITLE or FILENAME.

BEFORE (MENTION > SUBJECT)

Query: "What is Traefik?"

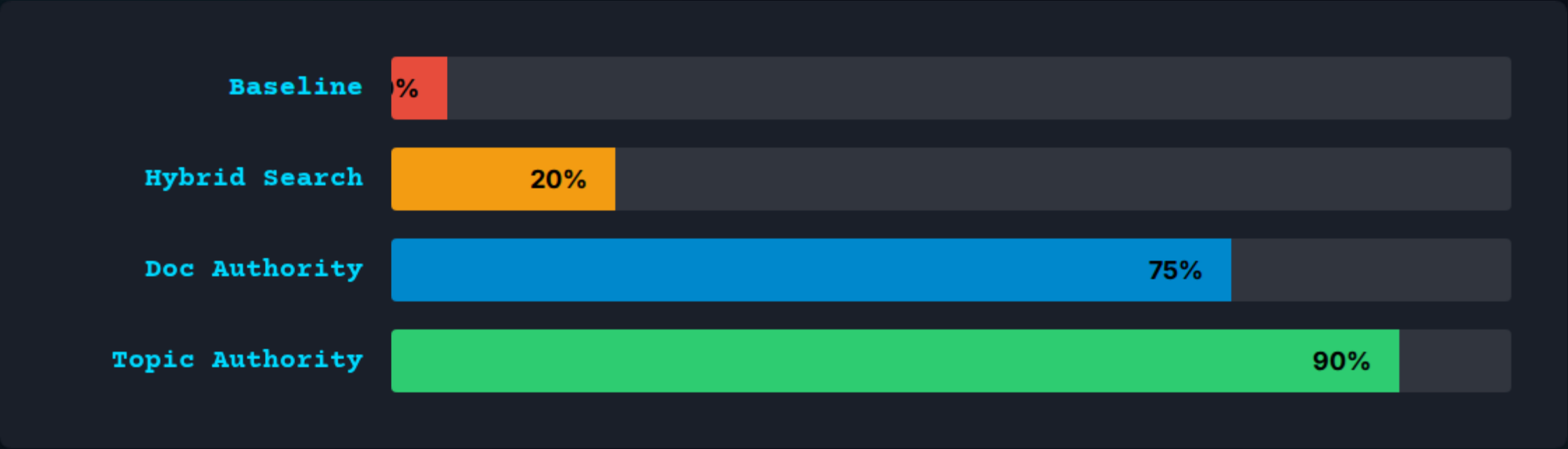
- ✗ ADR-002 mentions it 50 times (Rank 1)
- ✓ ADR-003 is ABOUT it (Rank 4)

AFTER (SUBJECT > MENTION)

- ADR-003 contains "Traefik" in title.
 - Receives 1.5× Topic Boost.
 - Rises to Rank #1.

RESULT: 75% → 90% accuracy across all categories.

THE TRANSFORMATION



FACTUAL 80% Correct	HOW-TO 80% Correct	ARCHITECTURAL 100% Perfect	COMPARATIVE 100% Perfect
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WORKFLOW DISCOVERY

INSIGHT: TOOL SELECTION MULTIPLIER

Session 6 revealed that matching the AI interface to the task tripled productivity. Different tools for different development stages.

CLAUDE CHAT

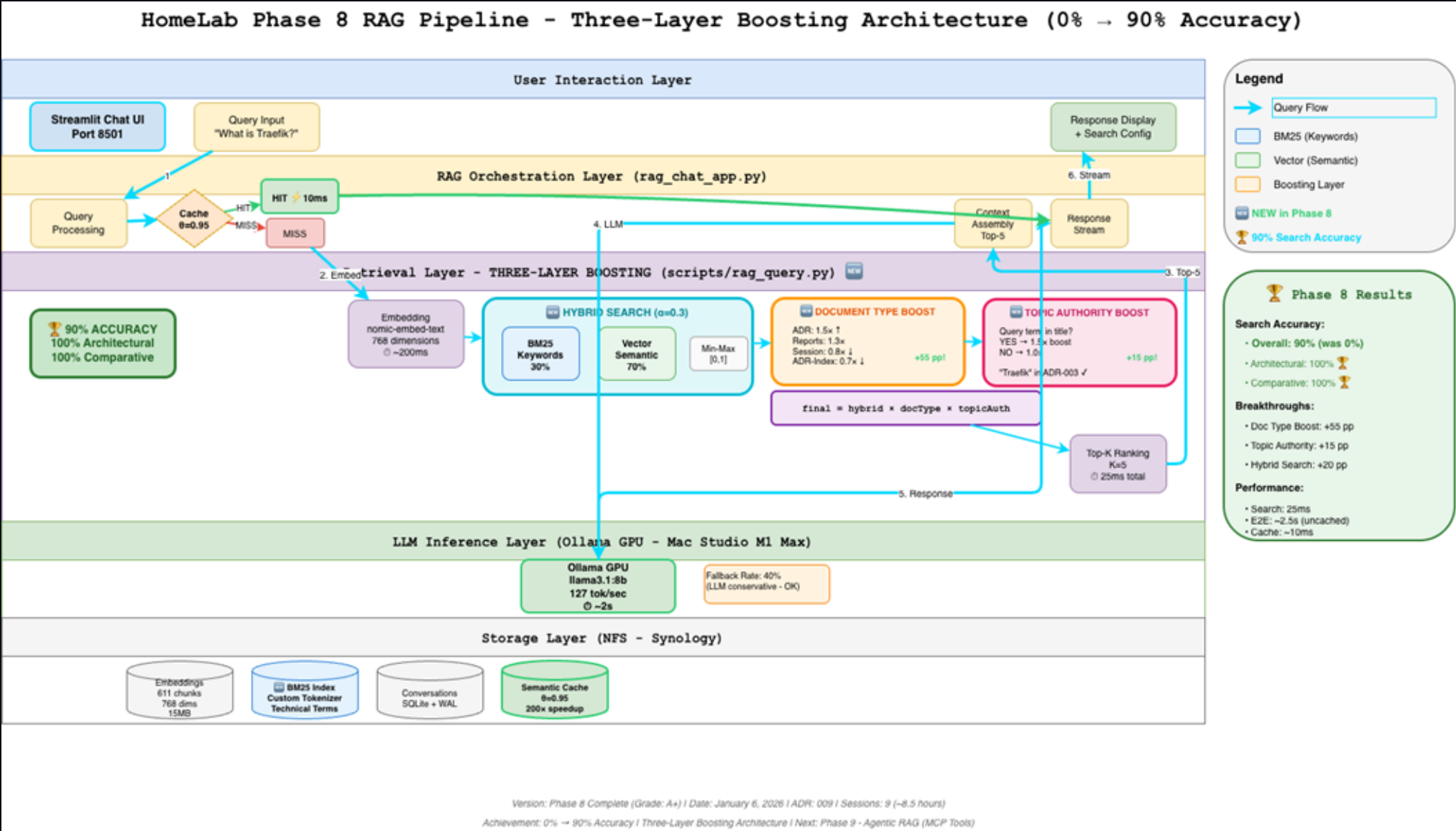
Best for planning, strategy, and context preservation.
Core for architectural decision-making and session logging.

CLAUDE CODE

Best for implementation, test validation, and git operations. Used for rapid re-ranking logic iteration.

Result: Topic Authority implemented in 20 minutes with zero regressions and perfect git history.

TECHNICAL ARCHITECTURE (1/2)



TECHNICAL ARCHITECTURE (2/2)

$$\text{Final Score} = (\text{Hybrid_Rank}) \times (\text{Doc_Type_Boost}) \times (\text{Topic_Match_Boost})$$



RETRIEVAL

Hybrid BM25 + Vector Search ($\alpha=0.3$). Min-Max Normalization ensures scores are mathematically comparable.



HEURISTIC LAYER

Multiplicative boosting for ADR types and Title matching. Heuristics solve what semantic math cannot reach.



LATENCY

Search Latency: 25ms average.
ADR Boost: 1.5× | Notes De-boost: 0.8× | Topic Match: 1.5×.

KEY ENGINEERING PRINCIPLES

1. MEASURE FIRST

Baseline: 0%. Diagnostic scripts revealed ADRs buried at rank #10. Every incremental change was validated against the test suite.

2. VALIDATE ASSUMPTIONS

Discovered chunk size was 10× larger than assumed (1847 chars). This discovery saved 4 hours of potential rework.

3. SIMPLE SOLUTIONS WIN

Topic Authority: 10 lines of code gave +15 percentage points. Elegance in logic beats complexity in math.

4. METHODOLOGY > MAGIC

Three pillars strategy. Each pillar solved one specific technical failure through rigour rather than luck.

90%

PHASE 8 READY

THE AGENTIC SHIFT

VISION FOR PHASE 9

From a static Knowledge Base to an Intelligent Operations Assistant.

- Live System Access: APIs for Nomad, Consul, and Prometheus.
- Intelligent Diagnosis: Knowledge Base vs. Live Cluster State
- Cross-Reference: Automate documented threshold compliance checking.

"The breakthrough wasn't better search. It was better ranking. Document authority matters more than pure similarity."

— Core Lesson from Phase 8

COMPOUND GAINS

Hybrid (+20), Doc Auth (+55), Topic Auth (+15). A multiplicative engineering journey.

SIMPLE IMPACT

Total boosting logic: <60 lines of code. Total Accuracy gain: 90 percentage points.

THE IMPACT

90%

ACCURAC
Y

100%

ARCH /
COMP

7.5

HOURS
INVESTED

"Engineering methodology transformed a broken search system
into elite-tier performance."

PHASE 8 COMPLETE | JANUARY 2026

APPENDIX

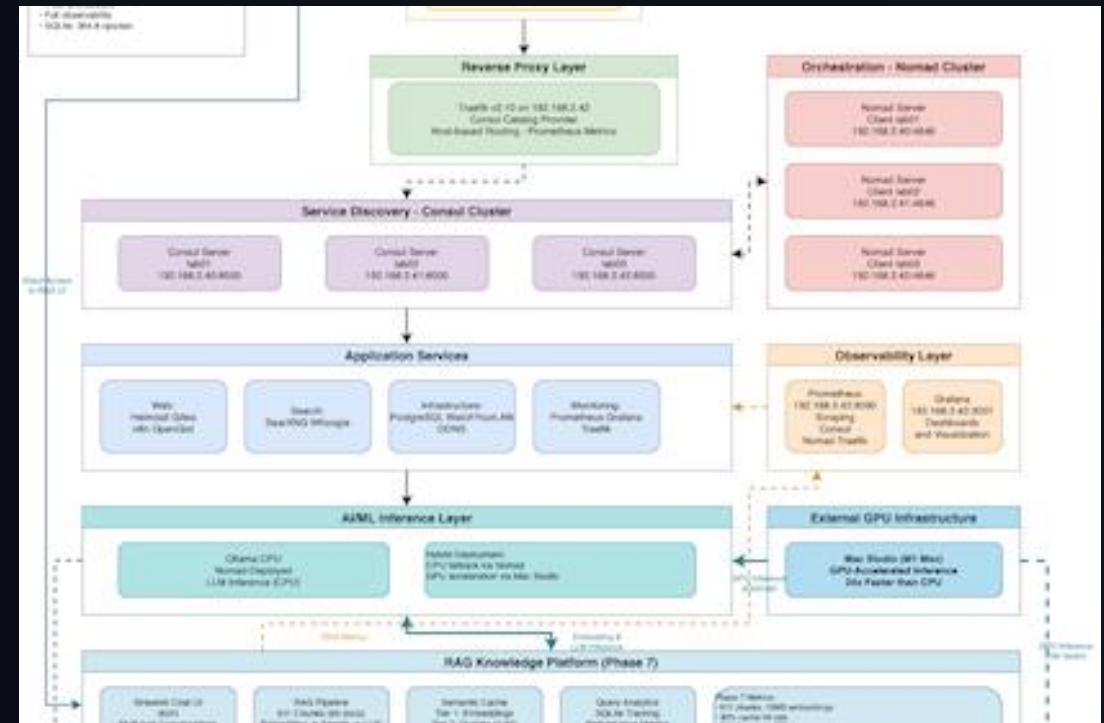
Detailed Architectural Diagrams

NOMAD CLUSTER ARCHITECTURE

SYSTEM TOPOLOGY

The Homelab environment operates on a multi-node cluster managed by HashiCorp Nomad. It ensures high availability for our search engine, storage backends, and UI components.

- Server Nodes: Governance & Scheduling
- Client Nodes: Workload Execution
- Consul Integration: Service Discovery



RAG PIPELINE ARCHITECTURE

THE RETRIEVAL FLOW

Our optimized pipeline integrates three distinct signal processing steps. It transforms raw vector similarity into authoritative context for the LLM generation step.

- Ingestion: 1847 char chunks
- Retrieval: Hybrid Alpha 0.3
- Re-Ranking: Authority + Topic Boost

